How Far Can Simulation Take Root Cause Diagnosis in Milling? Provenance-Tracked Simulation, Real-Data Gap Measurement and the Circularity of Simulated RCA Benchmarks

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Abstract

Root cause analysis (RCA) in machining aims to explain why a milling operation departs from healthy behaviour, for example because of a loose clamp, a worn tool, an excessive depth of cut or an unsuitable spindle speed, and it is a prerequisite for autonomous, zero-defect production. Real cuts with known causes are scarce and costly to produce, so simulation is the natural source of labelled data, and models trained on simulated cuts have already been transferred to real machines for single fault families such as chatter. These studies, however, build the diagnostic models and the test faults from the same simulator, treat one fault family on one machine, and do not measure how large the gap to real signals is or which part of it matters. We present a provenance-tracked regenerative-milling simulator whose healthy baseline is modelled from real multi-sensor spectra, and an evaluation framework that separates the healthy gap, the fault-signature gap and the circularity of the diagnostic procedure, applied to six causal RCA algorithms and to real-data transfer baselines. Experiments show that accuracy that is near perfect inside the generator falls from 0.80–0.98 to 0.35–0.71 when the generator's parameters are shifted, that the causal graph contributes nothing in this benchmark, and that no simulated fault reproduces the signature of a real defective tool. The simulator, the benchmark and the protocol are open source at `https://github.com/danghaidang04/RCA-for-Chatter-Simulation`.

1 Introduction

Milling is the workhorse of discrete manufacturing, and its quality depends on a small set of physical conditions that are easy to disturb: how stiffly the workpiece is clamped, how far the tool overhangs, how worn the cutting edge is, and whether the programmed depth of cut and spindle speed lie inside the stable region of the machine–tool system. *Root cause analysis* (RCA) for milling takes the signals that a machine already produces (accelerometers on the spindle, a microphone, spindle power and controller values, sometimes a force dynamometer) and returns a ranked list of the physical causes that explain an observed deviation from healthy cutting. The remedy depends on the cause: a loose fixture is re-clamped, a worn tool is replaced, an excessive depth is re-programmed, and a speed that falls into a lobe pocket is moved. Detecting that something is wrong is therefore not enough; autonomous and zero-defect production, and the reduction of scrap and unplanned downtime, require the machine to say *why*.

Why the problem is hard. The internal quantities that define the causes (stiffness, damping, cutting-force coefficients, tool wear) cannot be measured during cutting, and the same symptom, for

instance a louder and more broadband signal, can arise from several of them. Real labelled data are the obvious route to learning the mapping from signals to causes, but they are expensive and often impossible to collect in numbers: causes such as a deliberately loosened clamp, an overloaded tool or a chipped flute damage parts, tools and sometimes the machine, and each requires a controlled cutting campaign on every machine of interest. Public machining datasets therefore record sensor streams together with, at best, expert labels of normal or abnormal operation [13], or a single measured quantity such as flank wear [5], but not several known root causes.

What has been done. Four lines of work address parts of the problem. *Physics-based models* predict the stability of milling from the tool-tip transfer function and the cutting-force coefficients [2]; they say whether a cut will chatter, but they need quantities that are rarely known for a given machine and tool, and they do not infer causes from measured signals. *Data-driven monitoring* learns chatter or tool-wear detectors from measurements and, to cope with changes of machine and condition, uses transfer learning, domain-adversarial training and meta-learning [16, 23, 25], sometimes combined with physics or digital twins [8, 15, 24]; these detectors target one fault family and rely on real labels from the source machines. *Simulation-to-real learning* trains on simulated cuts and applies the model to a real machine: random forests trained on process simulations detect chatter [1] and tool-related faults [6], and domain randomisation, adversarial refinement and simulation-based inference are general tools for the same purpose [4, 21, 22]. *Causal root cause analysis* ranks the nodes of a causal graph by how much their mechanism changed between healthy and anomalous data [7, 10, 11, 14, 19, 20]; it was developed for software systems and has not, to our knowledge, been applied to machining signals with known causes.

The remaining challenge. Simulation-to-real studies in machining report that a model works on a real machine, but they do not tell the reader how much of the accuracy is due to the simulator itself. Three parts of the simulation gap are not separated in the literature. First, the *healthy baseline* of a simulator may differ from real signals, so that a classifier learns to recognise the simulator. Second, the simulated *fault signatures* may differ from real ones, so that learning the former does not diagnose the latter. Third, the *diagnostic procedure* may exploit the generator: when the nodes of a causal graph are estimated by regressors trained on labels produced by the very simulator that generates the test faults, a high accuracy is guaranteed by construction (Figure 1). Without separating the three, a high simulated accuracy cannot be interpreted, and a simulator whose parameters have no source can silently decide the outcome of a benchmark.

Our solution. We build a pipeline in which every ingredient is accountable to real data (Figure 2). The *simulator* is a regenerative milling model whose parameters carry a provenance tier (fixed, ranged, learned or assumed), so that unsupported assumptions are visible. The *healthy baseline* is modelled from real accelerometer and microphone spectra rather than simulated, so that the healthy gap can be measured directly with a classifier two-sample test; the simulator contributes only the *difference* between a faulty and a healthy run, scaled to real effect sizes. The *RCA stage* runs six causal algorithms on a 14-node graph and measures the circularity of the benchmark by shifting the generator's parameters at test time and by ablating the graph and the way the nodes are built. Finally, the simulated and real signals are compared on real data: a healthy baseline, a real defective tool, and a public multi-machine wear dataset on which simulation is pitted against domain-adaptation and one-feature baselines. The insight is that the question "does simulation support diagnosis?" has to be decomposed before it can be answered, and that each component can be measured with data that are publicly available today.

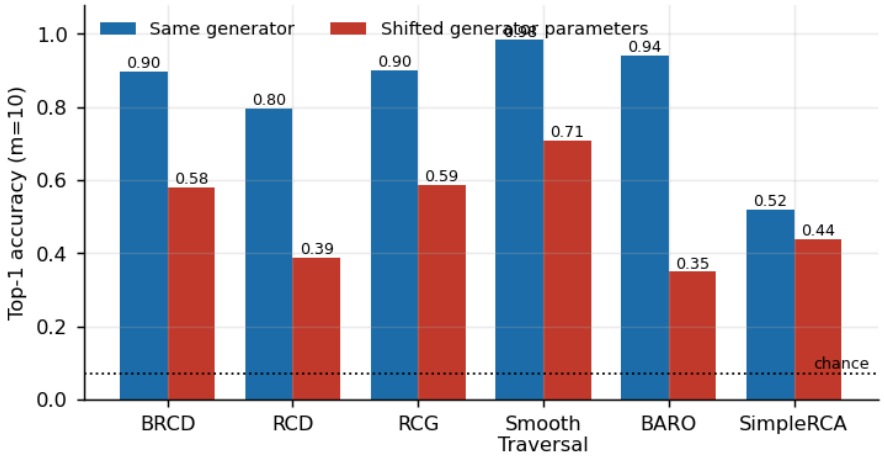


Figure 1: Why a simulated RCA benchmark cannot be read at face value: top-1 root-cause accuracy of six causal RCA algorithms (mean over five root causes, m=10 anomalous samples; dotted line: chance) when the diagnostic nodes and the test faults come from the same simulator (blue) and when the simulator's physical parameters are shifted at test time (red).

Findings. Accuracy near one inside the generator is not evidence of diagnostic ability: it drops sharply under a moderate shift of the generator, does not depend on the causal graph, and is close to chance when nodes are built without simulator labels. The simulated healthy baseline and the signature of a real defective tool remain distinguishable from real data, and the nominal simulator turns out to lie above its own stability limit for every real cutting condition. On a public wear dataset, simulation-only transfer to unseen machines is machine-dependent and does not beat a one-feature baseline on two of three machines, and fine-tuning from simulation is no better than training from scratch on this task.

Contributions.

1. **A measurement of benchmark circularity.** We show for six causal RCA algorithms that accuracy depends on how nodes are constructed and not on the causal graph, and quantify its collapse under generator shift (Section 4.2).
2. **A component-wise measurement of the simulation gap with negative results.** A real-data healthy baseline with a classifier two-sample test, a comparison with a real defective tool including a chipped-flute hypothesis, and a stability analysis of the simulator (Section 4.3).
3. **Evidence on when simulation helps on real data**, with domain-adaptation, one-feature and real-only baselines on a public three-machine dataset (Sections 4.2 and 4.4).
4. **An open pipeline**: a provenance-tracked simulator with FRF import, a precise account of the RCA algorithms as implemented and of their deviations from the originals, and an evaluation protocol for real multi-cause data (Sections 3 and 5.3).

2 Related Work

We organise the literature by the part of the problem it addresses and finish with a summary of what each line leaves open (Table 1).

2.1 Physics-based modelling of milling dynamics and chatter

Regenerative chatter is a self-excited vibration caused by the delay between the vibration marks left by successive teeth. Its stability can be predicted analytically with the zero-order approximation of the stability lobes, in the frequency domain with multi-frequency and semi-discretisation methods, and in the time domain with delay-differential-equation simulation [2]. These models take the tool-tip transfer function and the cutting-force coefficients as inputs and return a stability boundary or a simulated vibration. They are the right foundation for a simulator of faulty cuts because the effect of a stiffness, damping, force or speed change on the signal follows from the model. Their limitation for diagnosis is the inverse direction: they predict the signal from known parameters but do not infer the parameters (the causes) from measured signals, and the parameters of a given tool, holder and machine are rarely known precisely.

2.2 Data-driven monitoring and transfer for tool condition

Learning-based chatter detectors and tool-wear estimators are trained on measured vibration, sound, force or power signals. Because their accuracy falls when the machine, tool or cutting conditions change, transfer learning for chatter detection [25], domain-adversarial transfer for generalised tool-wear prediction [23] and semi-supervised multi-source meta domain generalisation across cutting conditions [16] have been proposed. Physics-guided deep learning [15], hybrid online data–model wear detection [24], digital twins with a dynamics core [8] and single-sensor wear prediction [9] add structure or reduce the sensing effort; systematic reviews of physics-informed learning for prognostics and health management and of simulation-to-real transfer in digital twins are given by Braun et al. [3] and Ma et al. [18]. These works address one fault family (chatter or wear), rely on real labelled data from the source domains, and do not rank several root causes. We use their ideas (domain-adversarial training, correlation alignment) as baselines on public wear data.

2.3 Simulation-to-real learning in machining

Alberts et al. [1] train a random forest on more than 140 000 simulated chatter cases and fine-tune it with 10% of 1 600 real instances from one custom three-axis machine, reaching 86.1% accuracy on a chatter-versus-stable decision; they state limitations related to the simulation-to-real difference. Ebrahimi Araghizad et al. [6] train random forests on analytical process-simulation data to monitor tool-related faults in milling and report 94% accuracy, and 93% on actual measurements (from the abstract; full text not read). These studies establish that simulation-to-real training is feasible for one fault family on one machine. They do not decompose the simulation gap, test how much of the accuracy comes from sharing the generator between training and testing, or consider several root causes and several machines. Generic tools for narrowing a simulation gap include calibration with a discrepancy term [12], domain randomisation [22], adversarial refinement of simulated data [21] and simulation-based inference [4]; we use a classifier two-sample test [17] as a diagnostic of the gap, not as a training signal.

2.4 Causal root cause analysis

Causal RCA formulates a failure as an intervention on one node of a causal graph and ranks nodes by the change of their mechanism between a healthy dataset and a few anomalous samples. RCD [10] tests the independence of a failure indicator from each metric under conditioning, RCG [11] ranks candidates by conditional mutual information given the parents in a partial causal structure, BARO [20] combines Bayesian change-point detection with a robust deviation score, SimpleRCA

Table 1: Coverage of the closest literature along the six aspects needed to judge simulation-based root cause diagnosis in machining. ✓: addressed; ○: partly; ×: not addressed; –: not reported or not applicable. ‡ judged from title and abstract only. C1 machining domain; C2 several root causes; C3 simulation used as the data source; C4 dependence on the generator tested; C5 gap to real signals measured; C6 parameter provenance reported.

Work	C1	C2	C3	C4	C5	C6
Chatter and stability theory [2]	✓	×	×	–	–	–
Chatter detection with transfer [25]‡	✓	×	×	–	–	–
Tool-wear transfer [16, 23]‡	✓	×	×	–	–	–
Physics-guided, hybrid, digital-twin wear [8, 15, 24]‡	✓	×	○	–	–	–
Simulation to real, chatter [1]	✓	×	✓	○	○	–
Simulation to real, tool faults [6]‡	✓	○	✓	–	–	–
Causal RCA, software [7, 10, 11, 19, 20]	×	✓	×	–	–	–
BRCD [14]	×	✓	○	○	–	–
This work	✓	✓	✓	✓	✓	✓

[7] compares the maximum of the anomalous data with the 95th percentile of the normal data, and Smooth Traversal [19] traverses the ancestors of a trigger node using outlier scores. BRCD [14] computes a Bayesian posterior over root causes from a CPDAG learned before the failure, with consistency guarantees, and compares ten methods on synthetic and microservice benchmarks; its authors observe that the methods based on marginal deviation (BARO, SimpleRCA) are competitive on two of the systems but do not generalise. These methods were developed and evaluated for software systems. Applying them to machining requires data with known causes, which is the role of simulation in this work; the algorithms in our code base are simplified re-implementations on a known graph (Section 3.4).

2.5 Datasets

The MSM dataset [13] provides synchronised accelerometer, microphone, current and controller data from three machines with expert labels (normal, abnormal, defective tool), and the LUH dataset [5] provides force and machine signals of three machines with three tools each and measured flank wear. Neither records several known root causes, which is why we use them for the healthy baseline, the signature of one real cause, and the study of transfer between machines.

2.6 Summary of the gaps

The table shows that no prior work combines machining, several root causes and a measurement of the simulation gap. The columns that are empty for all prior work (C4 and C6, partly C5) are exactly the ones needed to interpret a simulated benchmark, and they define the contributions of this paper.

3 Method

3.1 Problem formulation and overview

We consider a milling cut described by its condition $c = (n, a_e, a_p, f_t, \text{direction})$ (spindle speed, radial and axial depth, feed per tooth, up or down milling) on a machine–tool system θ (modal dynamics, force coefficients, sensor constants). A cut yields sensor signals s (two accelerometer

channels and a microphone, windowed to 0.3 s). A root cause is the change of one physical factor z_j among five (clamp stiffness, damping, tool wear, radial depth, spindle speed) relative to the healthy value. Given a healthy dataset D_{obs} and m anomalous windows D_{int} produced by an unknown cause, the *RCA task* is to return a ranking of the nodes of a causal graph G in which the true cause is as high as possible. Because real $(D_{\text{int}}, \text{cause})$ pairs are scarce, the data for developing and testing RCA come from a simulator that must be accountable to real data. Figure 2 gives the architecture; Section 3.2 describes the inputs, Section 3.3 the simulation and Section 3.4 the RCA stage.

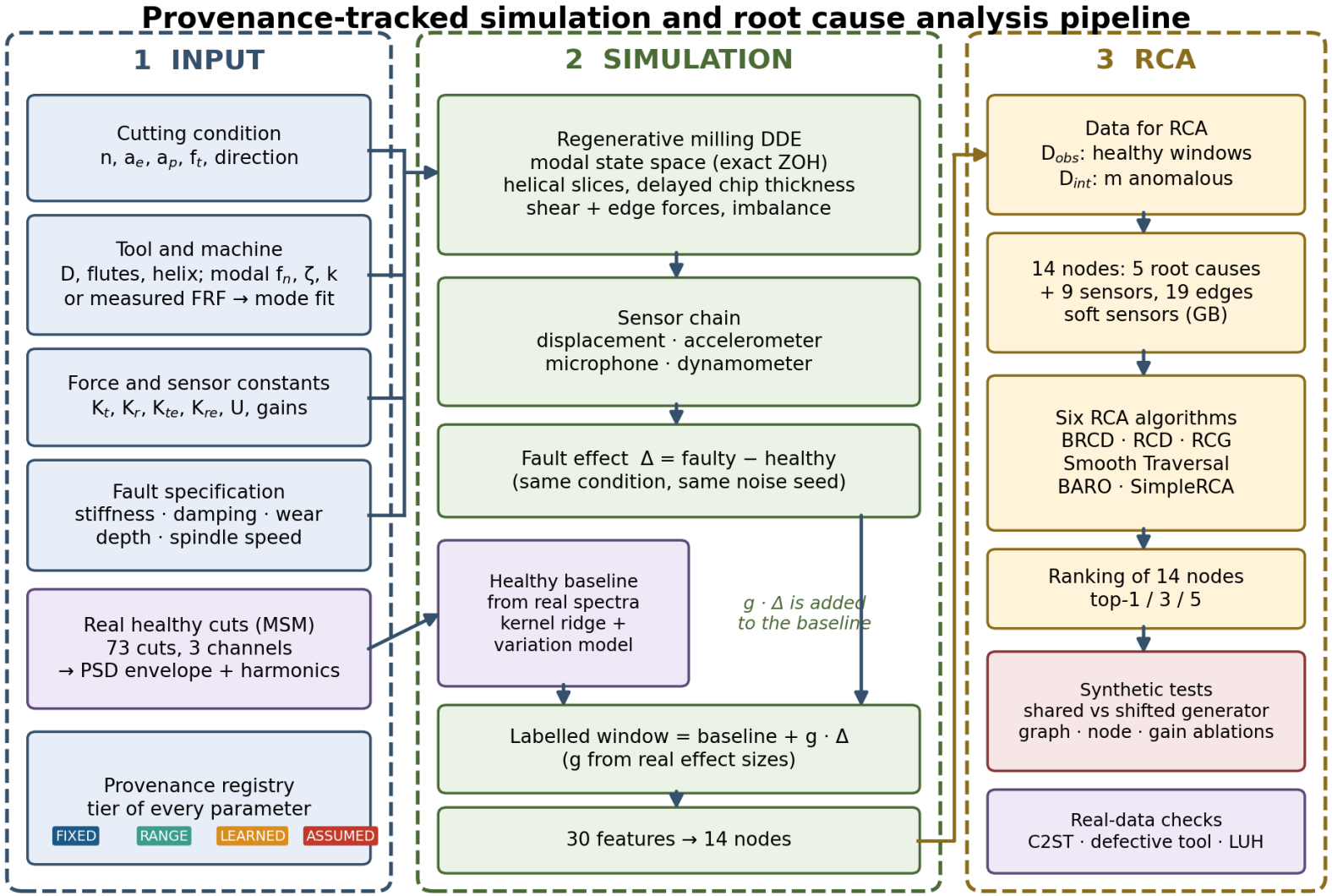


Figure 2: Architecture of the pipeline. **Input**: cutting condition, tool and machine, force and sensor constants, fault specification and real healthy cuts, all tagged in a provenance registry. **Simulation**: a regenerative milling model and a sensor chain give the fault effect Δ as the difference between a faulty and a healthy run; a baseline fitted to real spectra supplies the healthy signal. **RCA**: 30 features give 14 nodes, six causal algorithms rank them, and synthetic tests and real-data checks measure the three parts of the simulation gap.

Table 2: Input groups of the pipeline and their origin.

Input group	Content	Origin (provenance tier)
Cutting condition	n, a_e, a_p, f_t, direction, number of teeth	label files of MSM, checked against the controller (fixed)
Tool and machine	diameter, flutes, helix, flute length; modal f_n, ζ, k per axis	catalogue (fixed, helix medium confidence); modal values assumed, or measured FRF
Force and sensor constants	K_t, K_r, K_{te}, K_{re}, imbalance U, sensor gains and noise floors, runout	no verified source (assumed); learning failed (Section 4.3.8)
Fault specification	stiffness, damping, wear, depth, speed; severity ranges	assumed (no source)
Real healthy cuts	73 cuts, 219 windows of 0.3 s, three channels	MSM datasets 5 and 6 (real data)

3.2 Input

Motivation. The result of a simulation-based benchmark depends on every number that enters the simulator; the inputs therefore have to be explicit, and each must say where it comes from.

Design. The input consists of five groups (Table 2). (i) The *cutting condition* is read from the label files of the real data (spindle speed, feed, radial depth, direction) and verified against the controller values (ratio 1.000). (ii) The *tool and machine description* fixes the geometry (2SE0250IX150A: two flutes, 6.35 mm, flute length 38.1 mm, helix 30°) and the modal dynamics of the tool tip; for a new machine the dynamics are read from a hammer-test transfer function by the FRF importer (Section 4.3.7). (iii) The *force and sensor constants* (cutting-force coefficients, imbalance, gains and noise floors) are the parameters that had no usable source. (iv) The *fault specification* defines five faults and their severity ranges. (v) The *real healthy cuts* supply the spectra from which the healthy baseline is built. Every parameter is registered with a tier: *fixed* (a verified source gives the value for this exact tool and context), *range* (a source gives a range), *learned* (only data constrain it) or *assumed* (neither), with its context and verification status (Figure 3, Appendix A).

Advantage. The registry makes unsupported assumptions visible to the reader and prevents a literature value from being reused outside the context in which it was measured.

3.3 Simulation

3.3.1 Structural dynamics and cutting forces

Motivation. Regenerative chatter arises from the delayed feedback between the vibration marks left by one tooth and the chip thickness of the next; a time-domain model with the correct delay reproduces both stable forced vibration and saturated chatter, and the effect of a stiffness, damping, force or speed change.

Design. Each tool-tip axis $q \in \{x, y\}$ is a sum of modes with transfer function

$$H_q(s) = \sum_r \frac{\beta_r s + \alpha_r}{s^2 + 2\zeta_r \omega_r s + \omega_r^2}, \qquad \beta_r = 2\sigma_r, \quad \alpha_r = 2(\zeta_r \omega_r \sigma_r - \omega_{d,r} \nu_r), \tag{1}$$

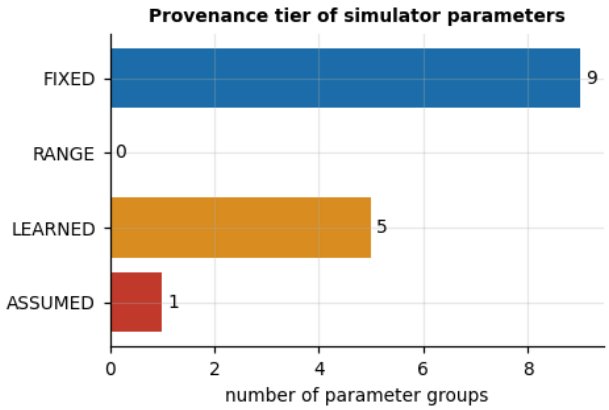


Figure 3: Provenance tiers of the simulator parameter groups.

(residues σ_r, ν_r; the nominal tool uses one mode per axis with $H = 1/[m(s^2+2\zeta\omega s+\omega^2)]$, $m = k/\omega^2$). Each mode is a two-state system $[x_r, \dot{x}_r]$ discretised exactly with a zero-order hold, (A_d, B_d) read from the matrix exponential of $\left(\begin{smallmatrix} A\Delta t & B\Delta t \\ 0 & 0 \end{smallmatrix}\right)$, at $\Delta t = 1/25\,600$ s. Tooth k of a flute with N teeth is divided into N_z=8 axial slices of height $dz = a_p/N_z$; the angle of slice q at time t is

$$\phi_{k,q}(t) = \Omega t + \frac{2\pi k}{N} - z_q \frac{2\tan\beta}{D}, \qquad z_q = (q + \tfrac{1}{2})dz, \tag{2}$$

where Ω is the spindle angular speed, β the helix angle and D the tool diameter. A slice cuts only while $\phi_{k,q} \in [\phi_{st}, \phi_{ex})$, with span $\arccos(1 - 2a_e/D)$ (up milling: $[0, \text{span})$; down milling: $[\pi - \text{span}, \pi)$). Its chip thickness depends on the current and delayed displacement,

$$h_{k,q} = \Big[f_t \sin\phi_{k,q} + \big(x(t) - x(t-T)\big)\sin\phi_{k,q} + \big(y(t) - y(t-T)\big)\cos\phi_{k,q} + \nu_k(t) \Big]_0^{h_{\max}}, \quad T = \frac{60}{N\,n}, \tag{3}$$

with n the spindle speed in rpm, f_t the feed per tooth (offset by $\pm$ the runout for the first two teeth), a fractional delay $T/\Delta t$ implemented by linear interpolation, tooth-wise roughness noise $\nu_k \sim \mathcal{N}(0, 0.02\,\mu\text{m})$, and $h_{\max} = 0.3$ mm. The elemental forces are

$$dF_t = (K_t h + K_{te})\,dz, \qquad dF_r = (K_r K_t h + K_{re})\,dz, \tag{4}$$

$$F_x \mathrel{+}= -dF_t\cos\phi - dF_r\sin\phi, \qquad F_y \mathrel{+}= dF_t\sin\phi - dF_r\cos\phi, \tag{5}$$

and spindle imbalance U adds $U\Omega^2[\cos\Omega t, \sin\Omega t]$.

Advantage and check. The core runs in numba (13 times faster than the Python reference). With helix and edge forces off it reproduces the pure-Python straight-flute reference to 7×10^{-16}, and eight slices are converged (Section 4.3.1).

3.3.2 Stability lobes

For comparison the zero-order approximation [2] gives, for each chatter frequency ω_c, the eigenvalue $\Lambda = -\frac{1}{2a_0}\big(a_1 - \sqrt{a_1^2 - 4a_0}\big)$ with $a_0 = H_x H_y(\alpha_{xx}\alpha_{yy} - \alpha_{xy}\alpha_{yx})$ and $a_1 = \alpha_{xx}H_x + \alpha_{yy}H_y$, the limit depth $a_{\lim} = -\frac{2\pi\Lambda_R}{NK_t}(1+\kappa^2)$, $\kappa = \Lambda_I/\Lambda_R$, and the spindle speeds $n = 60/(NT_k)$, $T_k = (\pi - 2\psi + 2k\pi)/\omega_c$, $\psi = \arctan(\Lambda_I/\Lambda_R)$. The lobes provide an independent check of the time-domain model and a test of whether a parameter set operates in the stable region.

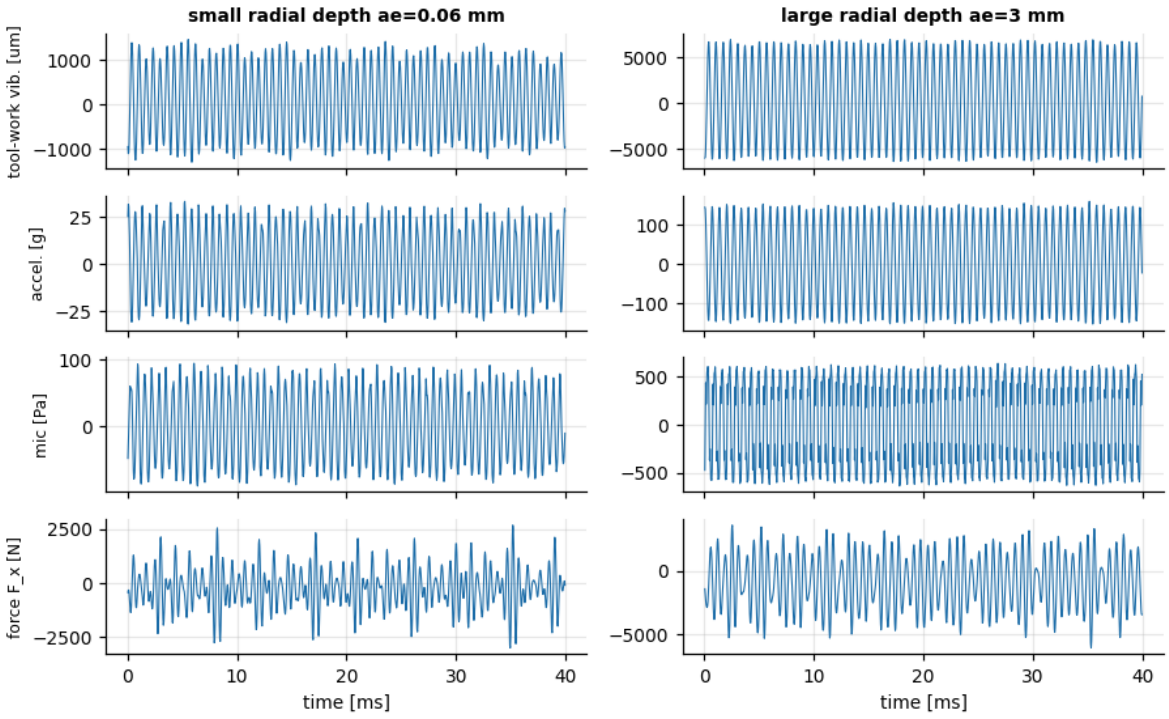


Figure 4: Simulated four-sensor signals at nominal parameters for the two extreme radial depths of the real data (10 000 rpm, 2-flute 6.35 mm tool, a_p=38.1 mm). The amplitudes (millimetre-scale displacement, tens to hundreds of g) are not physical for stable cutting; see Section 4.3.2.

3.3.3 Sensor chain

Motivation. Real RCA data are sensor signals, not displacements; the simulated displacement has to pass through the same kind of measurement chain.

Design. Four linear time-invariant blocks implemented in `scipy.signal`: (1) a relative tool–work displacement sensor (low-pass at 8 kHz, noise 0.05 μm); (2) an accelerometer, $a = \ddot{q}/g$ band-passed to 5 Hz–10 kHz with a gain to the sensor location and IEPE noise; (3) a microphone, $p = \frac{\rho_0 A}{4\pi r} a_{\text{res}} + c_{\text{cut}} \dot{F}_x$ delayed by r/c, band-passed to 20 Hz–10 kHz, plus floor noise; and (4) a force dynamometer, $-F$ through a second-order system (f_n=2 kHz, ζ=0.03) plus drift and noise (Figure 4). Simulation windows of 0.6 s are generated and the last 0.3 s analysed; signals are resampled with a polyphase filter to the sampling rates of the real data so that identical features can be computed.

3.3.4 Healthy baseline from real spectra

Motivation. Fitting the simulator's absolute signal levels directly to real windows did not converge (Section 4.3.8), and Section 4.3.2 explains why the nominal simulator cannot supply a realistic healthy signal; the data constrain the *healthy signal* far better than the physics constants.

Design. For each 0.3 s window we compute the power spectral density of two accelerometer channels and the microphone and summarise it by a piecewise-linear envelope over log-frequency (18

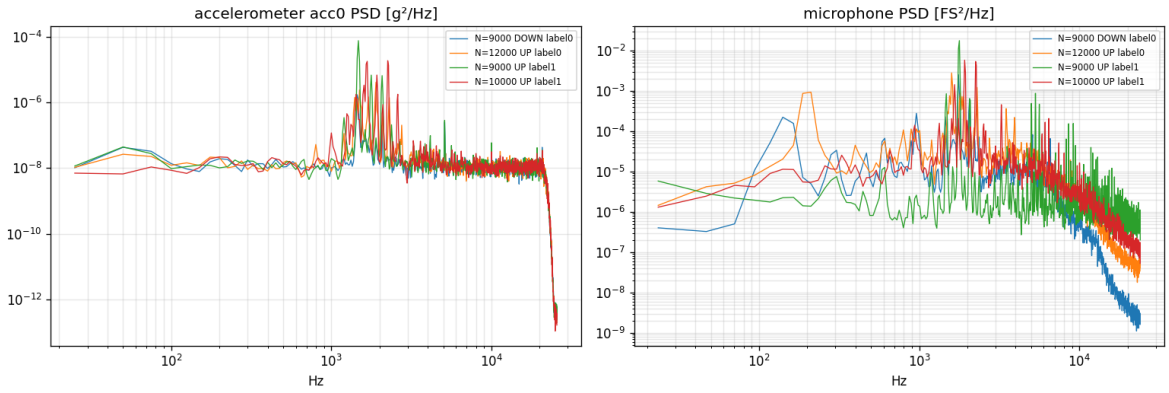


Figure 5: Power spectral density of four real windows (accelerometer left, microphone right): normal and abnormal cuts at different spindle speeds.

knots, band-averaged linear power so that energy is preserved) plus the rms of the first eight harmonics of the spindle frequency. Envelope parameters are regressed on the cutting conditions (kernel ridge on $\log n$, $\log a_e$, $\log f$, direction); out-of-fold residuals are split by one-way variance analysis into between-cut and within-cut parts on twelve principal directions. Windows are synthesised by shaping complex Gaussian spectra with the envelope and adding harmonics with random phases (real spectra in Figure 5).

Advantage. The healthy part of a simulated window is real-data-driven, so any remaining difference to real data (Section 4.3.3) is attributable to the baseline model, not to the physics.

3.3.5 Fault injection with effect-size gain

Motivation. A simulator's absolute fault amplitude is unknown, but the *direction* of a fault's effect on the signal is physically constrained.

Design. For each window and fault we run the simulator twice with the same condition and noise seed and add $g\,(s_{\text{fault}} - s_{\text{healthy}})$ to the baseline, with separate $g_{\text{acc}}, g_{\text{mic}}$. Faults and ranges (assumed): stiffness $\times[0.4, 0.9]$ (natural frequency $\propto \sqrt{k}$); damping $\times[0.3, 0.8]$; tool wear $K_t \times [1.1, 1.6]$ with $K_{te}, K_{re} \times (1 + 3(s-1))$; radial depth $\times[1.5, 4]$; spindle speed $\pm[3, 15]\%$. Matching feature distributions did not identify the gains (flat objective), so we set them by effect size: the mean shift of $\log_{10}$ acceleration rms and microphone level equals the real abnormal–normal difference (+0.375 and +4.86 dB), giving $g_{\text{acc}} = 7.7 \times 10^{-6}$ and $g_{\text{mic}} = 0.0116$.

Limit. Real abnormal windows are mainly up-milling cuts, so the scale is an assumption about fault strength, not an estimate (Section 4.3.5).

3.3.6 Features and nodes

Each window is described by 30 features: $\log_{10}$ acceleration rms, microphone level (dB), two spectral non-synchronous indices, ten log band powers each for the accelerometer and the microphone, the kurtosis of both, and the logarithms of spindle speed, radial depth and feed together with the milling direction. Nine of them are aggregated into the sensor nodes of Section 3.4.

Algorithm 1 Benchmark loop for one fault, condition and number of samples m.

Require: healthy windows $\mathcal{H}$, faulty windows $\mathcal{F}_c$ of cause c, soft sensors $\{g_j\}$, graph G, algorithm $\mathcal{A}$

$D_{\text{obs}} \leftarrow$ nodes of $\mathcal{H}$ (soft sensors out-of-fold), standardise with its mean and s.d.
for trial $= 1, \ldots, 100$ **do**
  draw m windows from $\mathcal{F}_c$ with replacement if needed; compute nodes with g_j; standardise $\rightarrow D_{\text{int}}$
  ranking $\leftarrow \mathcal{A}(D_{\text{obs}}, D_{\text{int}}, G)$
  record whether node(c) is at rank 1, within 3, within 5
end for
return mean top-1/3/5 over trials and over the five causes

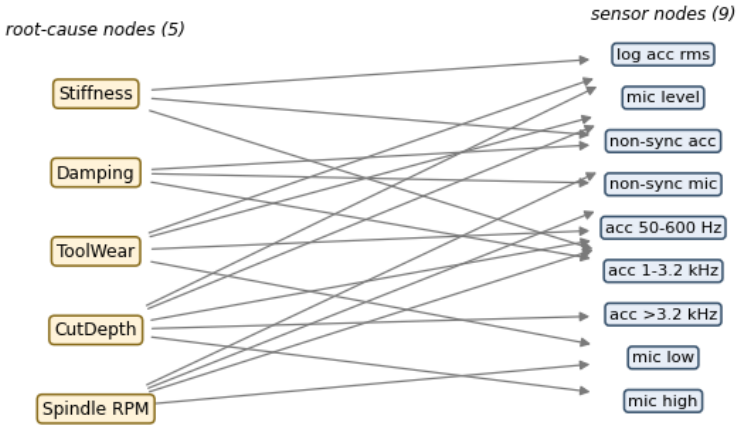


Figure 6: Causal graph used by the RCA algorithms: five root-cause nodes (left) and nine sensor nodes (right); 19 edges from physical prior knowledge.

3.4 Root cause analysis

3.4.1 Protocol

Following the root-cause-ranking protocol of Lee et al. [14], a healthy dataset D_{obs} and m anomalous samples D_{int} are given, the failure is treated as an intervention on one root-cause node, and each algorithm returns a ranking of the nodes; we report top-k accuracy of the true root-cause node (chance $k/14$). Cuts are split by identity into 43 training and 30 test cuts (stratified by a_e); each point averages 100 random draws of m faulty windows per fault, and all nodes are standardised with D_{obs} so that no algorithm benefits from scale differences (Algorithm 1).

3.4.2 Nodes and graph

The graph has 14 nodes (Figure 6, Table 3): five root-cause nodes and nine sensor nodes (log acceleration, microphone level, two non-synchronous indices, three accelerometer and two microphone bands). The 19 directed edges encode physical prior knowledge: each root cause points to the sensor quantities it can affect. Four root-cause nodes are soft sensors, i.e. gradient-boosted regressors of the log fault factor trained on training cuts only (out-of-fold predictions for D_{obs}); the spindle-speed node is the controller-measured deviation.

Table 3: Root-cause nodes.

Root-cause node	Source	Quantity	Construction
Stiffness_est	soft sensor	log stiffness factor	gradient boosting on 30 spectral features
Damping_est	soft sensor	log damping factor	same
ToolWear_est	soft sensor	log wear factor (K_t)	same
CutDepth_est	soft sensor	log radial-depth factor	same
SpindleRPM_est	controller	$\log(N/N_{\text{plan}})$	true value + 0.5% read noise

3.4.3 The algorithms as implemented

Table 4 compares each algorithm with its description in Lee et al. [14] (Appendix C of that paper) and states what our code (`run_full_paper_benchmark.py`) does. **The implementations are simplified, graph-aware re-implementations, not the authors' code**: all of them score a node by the standardised shift of its mean or of its residual given its parents, under a linear-Gaussian model, and BRCD, RCG, RCD and Smooth Traversal use the known graph. BARO's scoring step and SimpleRCA match their published descriptions. BRCD, RCG, RCD and Smooth Traversal contain constants with no source (0.12, 0.40, 0.70, 0.05). Consequently, our accuracies describe these re-implementations on this benchmark and must not be read as the performance of the original algorithms; the finding that the graph is irrelevant is partly a property of the re-implementations (Section 4.4).

3.4.4 Generator shift and real-data transfer

Generator shift. Because the soft sensors are trained on labels from the same generator, a *shifted* test multiplies natural frequency, damping, stiffness, force and edge coefficients by random factors in $[0.75, 1.33]$, $[0.5, 2]$, $[0.6, 1.7]$, $[0.6, 1.6]$ and $[0.4, 2.5]$ per cut, with soft sensors left at their nominal training.

Real-data transfer. For real wear data, features are expressed relative to the healthy baseline of each tool (the least-worn runs), are invariant to the sign convention of the force axes (absolute mean force) and are evaluated leave-one-machine-out; training modes are simulation only, simulation then fine-tuning on k real samples with an L2 penalty towards the simulated solution, and real only.

4 Experiments

4.1 Experimental setup

4.1.1 Configuration

All experiments run on a single Apple-silicon workstation with 10 worker processes; Table 5 lists the settings.

4.1.2 Datasets

Table 6 lists the real and simulated datasets. The simulated RCA set contains one window per class for each real window and each of three replicates ($219 \times 3 \times 6 = 3942$ windows), generated separately for a nominal and for a shifted generator; the simulated wear set contains 150 randomised machine–tool pairs with 45 runs each (6750 runs). In MSM datasets 5 and 6 the abnormal label is

Table 4: The six RCA algorithms: original method (as summarised by Lee et al. 14) versus our implementation.

Algorithm	Original method	Our implementation	Fidelity
BRCD [14]	Bayesian posterior over root causes from a CPDAG learned pre-failure, uniform DAG sampling per interventional Markov equivalence class; consistency guarantees	Linear-Gaussian local model per node on a given DAG; score = standardised mean shift of the node (root) or of its residual given parents (child), $\times(1 + 0.12\,\lvert\text{descendants}\rvert)$	heuristic; no DAG sampling
RCD [10]	random partitions; conditional-independence tests of the failure indicator F against each metric; rank by p-values	partial correlation test of F and the node given its parents in the known DAG; score $-\log_{10} p$	simplified; uses the true DAG
RCG [11]	marginal tests with F, then conditional mutual information given possible parents in a CPDAG	squared standardised shift of the residual given parents (Gaussian proxy for CMI)	simplified proxy
Smooth Traversal [19]	outlier score of the anomalous sample; traverse from the trigger to ancestors; largest score difference between a node and its parents	mean $\lvert z\rvert$ score; $z_i - 0.70 \max_{p\in\mathrm{pa}(i)} z_p$, plus $0.05\, z_i$	heuristic constants
BARO [20]	Bayesian online change-point detection of anomaly time; rank by median/IQR-scaled deviation	median/IQR-scaled deviation with the anomaly time known	scoring step matches
SimpleRCA [7]	95th percentile of normal data versus maximum of anomalous data	same	matches

strongly associated with up-milling (all 36 down-milling cuts are normal), so it is not a specific root cause.

4.1.3 Metrics

4.1.4 Baselines

The comparisons use five kinds of baseline (Table 8): the six RCA algorithms (re-implementations); direct supervised classifiers on the simulated windows as a within-simulator reference that ignores the graph; a chance rule and label-free nodes as lower references; for real wear transfer, a one-feature force rule without labels, standard classifiers, correlation alignment and domain-adversarial training with unlabelled target data, and real-only training; and a simulation-only model with and without fine-tuning.

4.2 Main results

4.2.1 Causal RCA on simulated faults: is the benchmark circular?

Protocol. We compare the shared-generator and the shifted setting for $m \in \{5, 10, 20, 50\}$ (Table 9, Figures 7 and 8).

Result. With a shared generator, Smooth Traversal and BARO approach 1.0 as m grows, BRCD and RCG reach 0.85–0.90, RCD declines with m, and SimpleRCA stays near 0.5. Under shifted

Table 5: Experimental configuration.

Component	Setting
Software	Python 3.13.12; numpy 2.4.4, scipy 1.18.0, scikit-learn 1.9.1, numba 0.68.0, torch 2.12.1
Simulator	$\Delta t = 1/25\,600$ s; window 0.6 s, last 0.3 s analysed; N_z=8; $h_{\max}$=0.3 mm; roughness noise 0.02 μm; tool 2SE0250IX150A (2 flutes, 6.35 mm, β=30°), a_p=38.1 mm; nominal f_n=1500 Hz, ζ=0.03, k=3 × 10^6 N/m, K_t=7 × 10^8 N/m^2 (assumed)
Healthy baseline	18 spectral knots, 8 harmonics; kernel ridge (RBF, α=0.3, γ=0.15) on five condition variables; ridge λ=1; 12 principal directions; C2ST by random forest with 3 stratified splits (22 test cuts)
Fault injection	severity ranges of Section 3.3.5; $g_{\text{acc}} = 7.7\times10^{-6}$, $g_{\text{mic}} = 0.0116$ (effect-size calibration)
RCA benchmark	14 nodes, 19 edges; 43 training and 30 test cuts; soft sensors HistGradientBoosting (150 iterations, learning rate 0.1, 5-fold grouped out-of-fold); $m \in \{5, 10, 20, 50\}$; 100 draws per fault; ablations: 3 seeds, 60 draws
Real wear transfer	LUH 406 runs; GB (150 iterations, rate 0.1), LR (C=0.3), CORAL (ϵ=1), DANN (hidden 64–32, dropout 0.2, 200 epochs, Adam 10^{-3}, weight decay 10^{-3}, domain-loss weight 0.5); 10 seeds; fine-tuning $\lambda \in \{0.3, 1, 3\}$, $k \in \{2, \ldots, 32\}$, 40 draws

Table 6: Datasets.

Dataset	Type	Content and use
MSM 5, 6 [13]	real	73 cuts (48 normal, 25 abnormal), 219 windows; 2-flute 6.35 mm tool, AA 6061 (wet); healthy baseline, gain calibration, C2ST
MSM 3, 14	real	defective 12.7 mm four-flute tool D* (12 cuts, 36 windows) and healthy tool (64 cuts, 192 windows), AISI 4140; fault signature
LUH [5]	real	3 machines × 3 tools, 406 runs, force at 25 kHz, flank wear VB; 26% of runs have VB $\geq$ 100 μm
Simulated RCA set	simulated	3942 windows per generator (nominal and shifted), 30 features, 14 nodes
Simulated wear set	simulated	6750 runs of randomised four-flute machine–tool pairs with growing wear

parameters all methods drop and the ordering changes: BARO falls from 0.94 to 0.35 at m=10, BRCD and RCG improve slowly with m, and Smooth Traversal remains best at 0.71 (top-3: 0.98 Smooth Traversal, 0.93 BARO, 0.89 BRCD). The accuracy is unevenly spread over the causes (Figure 8). Several entries equal 1.00 with a shared generator; we do not regard them as evidence of diagnostic ability, because the spindle-speed node contains the answer by construction and the other root-cause nodes are supervised estimates of it.

4.2.2 Does simulation help on real machines? Cross-machine wear

Protocol. LUH has three machines and three tools per machine with measured flank wear. The target is wear $\geq$ 100 μm, evaluated leaving one machine out (Table 10, Figure 9); a model trained only on simulated wear is applied without real labels.

Result. Wear detection with real labels is easy: gradient boosting with raw features is $\geq$ 0.976 on all three target machines. The one-feature rule reaches 0.996 and 0.993 on machines 1 and 2 but 0.46 on machine 3, so force scale differs across machines. The model trained only on simulated wear reaches 0.887, 0.807 and 0.571 using sign-invariant tool-relative features, which is below the one-

Table 7: Metrics. ↑ higher is better, ↓ lower is better.

Metric	Definition	Used for
Top-k accuracy ↑	true root-cause node within the first k ranks (chance $k/14$)	RCA benchmark
Macro-F1, recall per cause ↑	six-class classification of simulated windows	classifier reference
C2ST balanced accuracy ↓	random-forest accuracy separating real from simulated healthy windows; 0.5 means indistinguishable	healthy gap
Shift in s.d. and cosine	node shift of anomalous windows in healthy standard deviations; cosine of shift vectors	fault-signature gap
AUROC ↑	wear detection, VB $\geq$ threshold; chance 0.5	real wear transfer
MAE of VB ↓	error of wear estimation in μm	real wear transfer
Spearman ρ	rank correlation of proxy distance and AUROC	dynamics distance
ECE, coverage–accuracy	calibration and abstention	planned real multi-cause data

Table 8: Baselines and their role.

Task	Baselines	Role
Causal RCA	BRCD, RCD, RCG, Smooth Traversal, BARO, SimpleRCA (re-implementations); chance 1/14	compare ranking methods
Simulated classification	gradient boosting, random forest, QDA, logistic regression	supervised reference inside the generator
Node construction	supervised soft sensors versus label-free spectral nodes	measure dependence on simulator labels
Real wear transfer	one-feature RMS rule (no labels); GB; LR; CORAL + LR; DANN	non-simulation baselines
Simulation to real	simulation only; simulation + fine-tuning; real only (k runs)	benefit of simulation and of fine-tuning

feature rule on machines 1 and 2. The first attempt with signed mean force was below chance (0.55, 0.26, 0.10) because of an axis sign convention; changing to absolute mean force was decided after seeing this result, though with a physical argument, so these numbers should be read as exploratory. Bootstrap intervals are over runs and understate uncertainty because runs of one tool are correlated; gradient boosting and logistic regression are deterministic, so their seed spread is zero.

4.3 Analysis: is the simulator right?

4.3.1 Numerical correctness and stability lobes

With helix and edge forces off, the numba core equals the pure-Python straight-flute reference to 7×10^{-16}; eight axial slices are converged; and a time-domain grid agrees with the zero-order lobe prediction on 79% of speed–depth points, with disagreement at moderate depth where the zero-order approximation is known to be conservative (Figure 10).

Table 9: Causal RCA: top-1 accuracy ↑ (mean over 5 root causes; 100 draws per fault) versus the number of anomalous samples m; best per column in bold.

	Same generator				Shifted generator parameters			
Algorithm	m=5	10	20	50	m=5	10	20	50
BRCD	0.87	0.90	0.89	0.85	0.59	0.58	0.62	0.64
RCD	0.79	0.80	0.71	0.59	0.45	0.39	0.30	0.21
RCG	0.87	0.90	0.89	0.85	0.58	0.59	0.63	0.64
Smooth Traversal	**0.95**	**0.98**	**1.00**	**1.00**	**0.68**	**0.71**	**0.77**	**0.81**
BARO	0.88	0.94	0.98	1.00	0.42	0.35	0.30	0.26
SimpleRCA	0.53	0.52	0.52	0.48	0.42	0.44	0.44	0.40
Chance (1/14)	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07

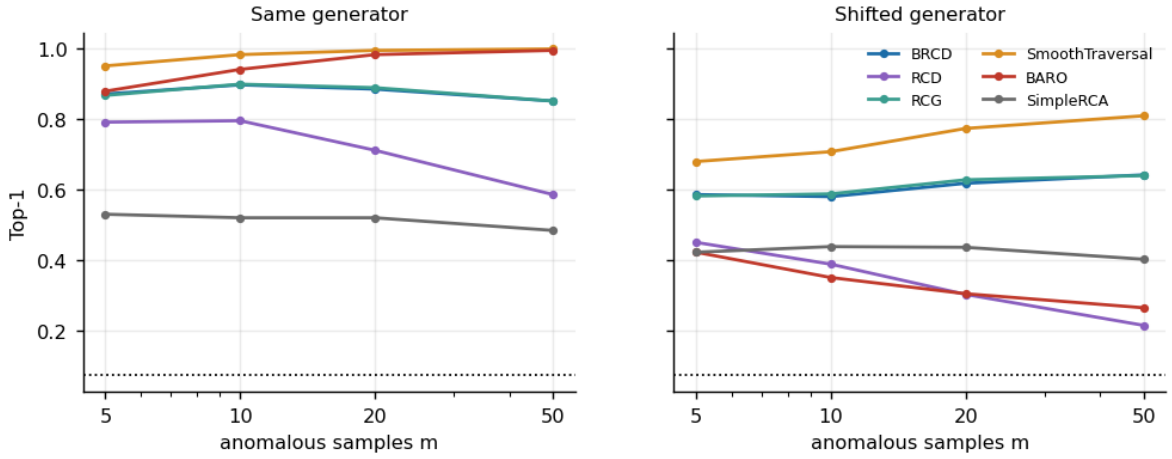


Figure 7: Top-1 accuracy versus the number of anomalous samples in the shared-generator and shifted settings (dotted: chance). In the shifted setting BARO and RCD deteriorate with more samples while Smooth Traversal improves.

4.3.2 Operating regime and robustness

Plotting the raw simulator output (Figure 4) showed millimetre-scale displacement. At nominal parameters (modal stiffness 3×10^6 N/m, f_n=1500 Hz, ζ=0.03, K_t=7×10^8 N/m^2) the zero-order limit depth at 9000 rpm is 30.6 mm for a_e=0.0635 mm and 18.0 mm for a_e=0.3175 mm, below the real axial depth of 38.1 mm; the time-domain rms displacement is 0.44 and 0.82 mm and the tool leaves the cut ($h_{\min} = 0$). Increasing the modal stiffness 3, 10 and 30 times moves the limit above the depth and the rms displacement to 36–159, 5–8 and $\leq$1 μm (Table 11). Real healthy cuts are presumably stable, so the nominal simulation is not an operating point of the real machine. Because the simulator only supplies fault *differences*, rescaled by gains chosen to match real effect sizes, the gains absorb the amplitude mismatch; with the modal stiffness multiplied by 10 the calibration needs gains 45 and 38 times larger ($g_{\text{acc}} = 3.5 \times 10^{-4}$, $g_{\text{mic}} = 0.43$; the search grid had to be extended). We repeated the causal benchmark in this stable regime (Table 12, Figure 11): top-1 accuracy at m=10 is 0.53–0.98 with a shared generator and falls to 0.32–0.62 under shifted parameters (nominal: 0.52–0.98 and 0.35–0.71). The ordering of the algorithms is partly regime-dependent: BARO no longer collapses under shift (0.50 instead of 0.35) and Smooth Traversal remains best (0.62). The graph ablation and the real-data comparisons were not repeated in this regime.

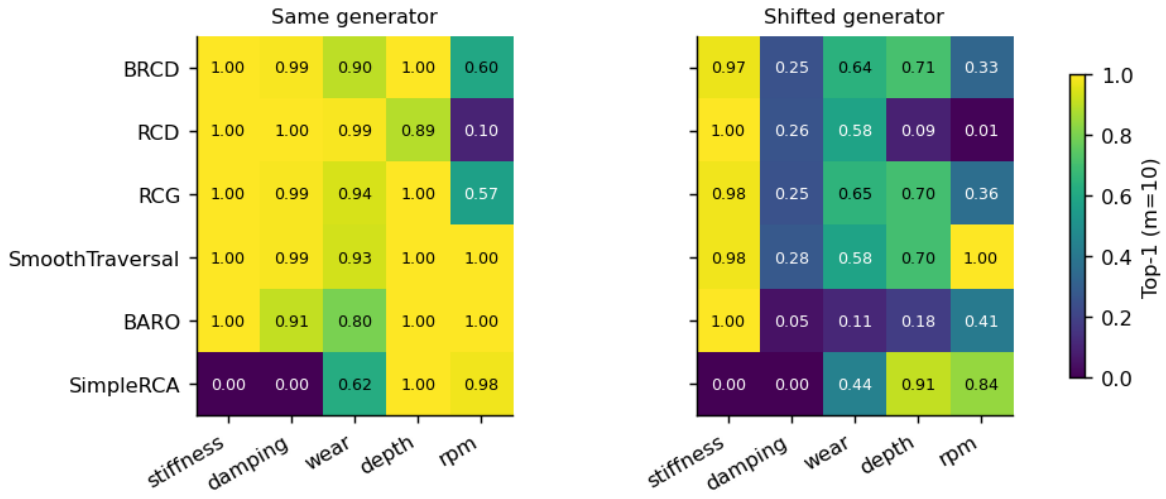


Figure 8: Top-1 accuracy per root cause and algorithm at m=10.

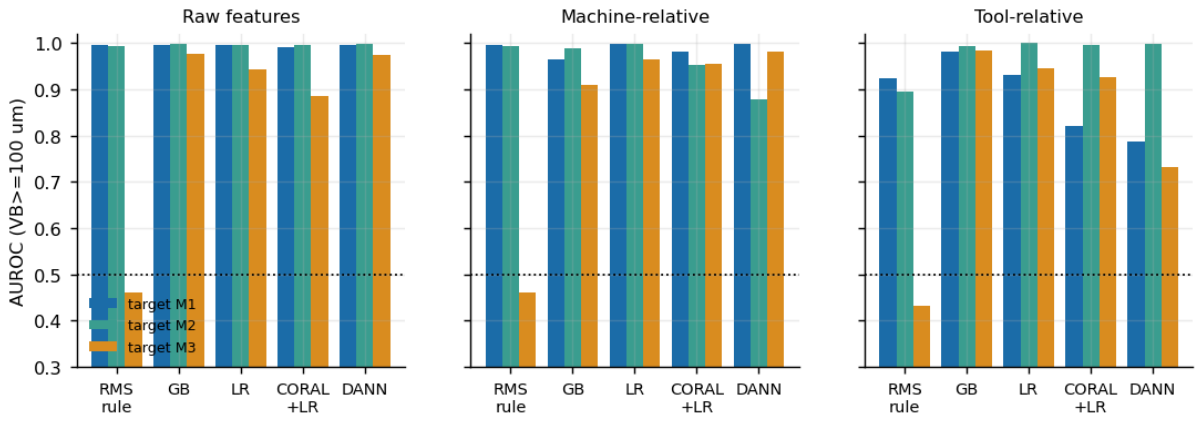


Figure 9: Leave-one-machine-out AUROC for three feature sets and five methods (dotted: chance).

4.3.3 How far is the simulated healthy baseline from real data?

We train the baseline on a subset of cuts and test on held-out cuts with a random-forest classifier two-sample test. Table 13 and Figure 12 show that all variants remain far from 0.5 (0.82 ± 0.04 for the final model) and that the kernel and hierarchical improvements are small relative to the split spread. Feature means match (log acceleration -1.37 simulated, -1.38 real) but dispersion and joint dependence on conditions do not. We could not measure a floor because no cutting condition is repeated. An earlier version used plain accuracy with five synthetic windows per real one (chance 0.83); those numbers are discarded.

4.3.4 Does any simulated fault reproduce a real defective tool?

Protocol. We compare the intentionally defective tool D* (12 cuts, 36 windows) with the healthy tool of the same type (64 cuts, 192 windows). Conditions differ, so we fit ridge models of each sensor node on the conditions with healthy cuts only (out-of-sample over cuts), express defective windows as residuals in healthy standard deviations, and bootstrap cuts for 95% intervals (Table 14, Figure 13).

Table 10: Leave-one-machine-out wear detection (VB ≥ 100 μm), AUROC ↑, mean over 10 seeds on target machines M1–M3.

Method (target machine →)	M1 ↑	M2 ↑	M3 ↑
Raw features			
One-feature RMS rule (no labels)	0.996	0.993	0.462
Gradient boosting	0.995	**0.999**	**0.976**
Logistic regression	0.996	0.995	0.943
CORAL + LR†	0.992	0.995	0.886
DANN†	**0.996**	0.997	0.974
Machine-relative features			
One-feature RMS rule (no labels)	0.996	0.993	0.462
Gradient boosting	0.965	0.989	0.910
Logistic regression	**0.998**	**0.999**	0.964
CORAL + LR†	0.981	0.952	0.956
DANN†	0.997	0.879	**0.980**
Tool-relative features			
One-feature RMS rule (no labels)	0.925	0.894	0.433
Gradient boosting	**0.980**	0.994	**0.985**
Logistic regression	0.931	**0.999**	0.946
CORAL + LR†	0.822	0.997	0.926
DANN†	0.786	0.998	0.731
Simulation only, no real labels (tool-relative x, y)	0.887	0.807	0.571

† uses unlabelled target-machine features (transductive).

Result. The real tool raises all levels, most at low frequency (+3.9 and +4.0 for the 50–600 Hz accelerometer and microphone bands) and in the 1–3.2 kHz accelerometer band (+6.3), and lowers both non-synchronous indices (-3.7, -1.2), i.e. the spectrum becomes more tonal. No simulated fault shifts low frequencies (all within ± 0.1) or the non-synchronous indices beyond -0.3 to $+0.5$. The cosine similarity to the five simulated shifts is 0.55–0.62 and driven by the common "louder" direction. The tools differ (12.7 mm four-flute steel versus simulated 6.35 mm two-flute aluminium), so only direction and sign in standardised units are compared.

Tested hypothesis: chipped flute. We added effective runout of 10–80 μm (a once-per-revolution term) and tried fault gains of 1×, 10×, 100× (Figure 14). The cosine similarity stays at 0.48–0.54, again dominated by overall level; non-synchronous indices move in the wrong direction (+0.1 to +0.9 versus -3.7 and -1.2), the low bands barely change (microphone low band $\leq +0.9$ versus +4.0), and the high accelerometer band grows far too much with added gain (+1 to +38 versus +1.9). A once-per-revolution runout therefore does not reproduce the signature either. Per-tooth force asymmetry and nonlinear edge or friction effects remain untested.

4.3.5 Sensitivity to the assumed fault strength

Scaling the fault gains by 0.3 and 3 changes the best simulated classifier's accuracy from 0.698 to 0.943 (Table 15, Figure 15); six-class accuracy within the simulator (Table 16, 16 held-out cuts) is therefore only interpretable together with this curve. Applied to real windows, 91–97% of normal windows are classified normal by tree and neighbour models but only 51% and 57% by logistic regression and Naive Bayes, and 61, 64 and 72 of 75 real abnormal windows are assigned to tool

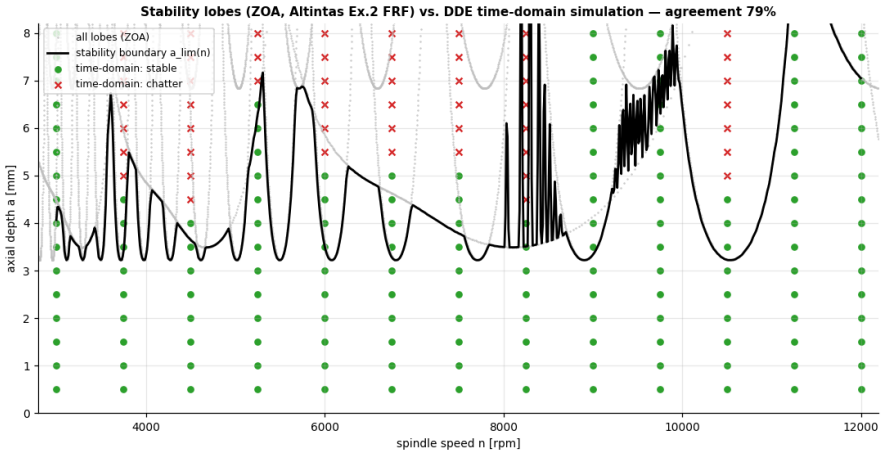


Figure 10: Stability lobes (zero-order approximation, 4-flute reference transfer function) and time-domain simulation outcomes on a speed–depth grid.

Table 11: Operating regime of the simulator at 9000 rpm, $a_p = 38.1$ mm, as a function of the modal stiffness scale (nominal = 1); red: limit depth below a_p. Source: `regime_check.py`.

Stiffness scale	a_e [mm]	ZOA $a_{\lim}$ [mm] ↑	x rms [μm] ↓	F_x rms [N]
1	0.0635	30.6	439.7	425
1	0.3175	18.0	821.8	659
3	0.0635	91.7	36.4	105
3	0.3175	53.7	159.1	387
10	0.0635	195.1	4.7	56
10	0.3175	179.8	8.0	76
30	0.0635	> 200	0.7	52
30	0.3175	> 200	0.9	58

wear by QDA, random forest and gradient boosting; with no ground truth for abnormal windows we read this as mapping any broadband increase to wear.

4.3.6 Does real-data transfer degrade when the target dynamics are far from the training machines?

Without measured transfer functions for LUH we used a proxy: the mean 21-band force spectrum of the three least-worn runs of each tool. For each of the nine target tools we trained on the tools of the other machines (all six, the three nearest, the three farthest by the proxy). With gradient boosting and raw features at VB ≥ 100 μm, AUROC was 0.993, 0.984 and 0.976 for all, nearest and farthest training tools; the Spearman correlation between proxy distance and AUROC was -0.16 ($p = 0.67$) at VB ≥ 100 μm and -0.34 ($p = 0.36$) at 60 μm (Figure 16). Only logistic regression on tool-relative features at 60 μm showed a trend ($\rho = -0.62$, $p = 0.08$; lowest AUROC 0.25). With nine targets, a weak proxy and an unknown range of dynamics, this is absence of evidence, not evidence of absence: on these data real-data transfer is sufficient, and whether simulation is needed to reach dynamics outside the training range remains open.

Table 12: Top-1 accuracy ↑ at m=10 (mean over 5 root causes) at nominal parameters and with the modal stiffness multiplied by 10; best per column in bold.

	Nominal parameters (above stability limit)		Stiffness ×10 (stable regime)	
Algorithm	Same generator	Shifted	Same generator	Shifted
BRCD	0.90	0.58	0.94	0.49
RCD	0.80	0.39	0.76	0.40
RCG	0.90	0.59	0.92	0.48
Smooth Traversal	**0.98**	**0.71**	**0.98**	**0.62**
BARO	0.94	0.35	0.94	0.50
SimpleRCA	0.52	0.44	0.53	0.32
Chance (1/14)	0.07	0.07	0.07	0.07

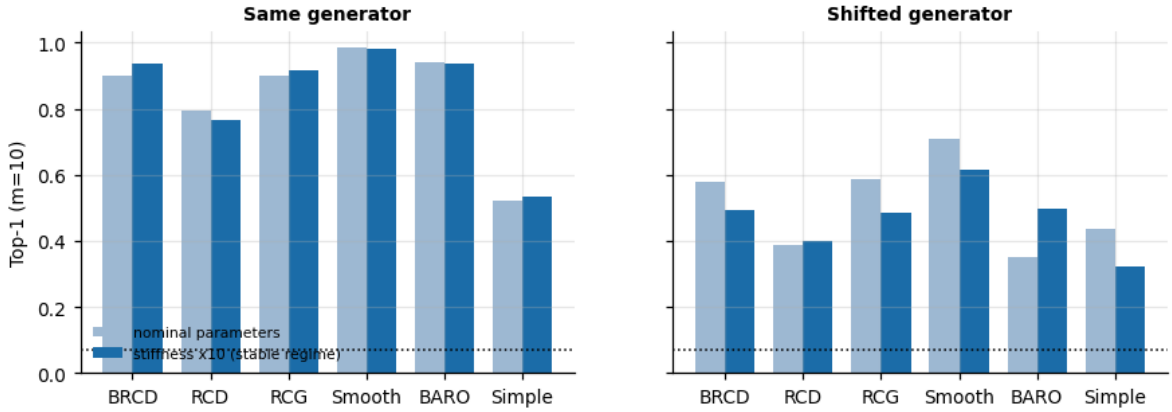


Figure 11: Robustness to the operating regime: top-1 at m=10 for nominal parameters and for a stiffer, stable-regime parameter set.

4.3.7 FRF import

For a new machine the tool-tip transfer function can be read from hammer-test data (receptance, or accelerance divided by $(j\omega)^2$) and fitted by adding modes at residual peaks with linear residue estimation. On synthetic transfer functions with added noise the shape error is 0.8% (no noise), 1.7% (noise 1% of the peak) and 16% (3%, one mode lost); the importer has not been tested on measured FRFs (Figure 17).

4.3.8 Negative result: direct calibration of the delay model

Fitting 15 physical and sensor parameters to real windows with CMA-ES (MMD, later group means over conditions) plateaued at a loss near 3 (squared errors in real standard deviations) and did not converge in 35–40 generations; we stopped by a pre-set rule. Earlier low-MMD fits had wrong dependence on conditions: real microphone level correlates with spindle speed at +0.85, the simulation gave 0.00; simulated acceleration depended on radial depth (0.69), the real one did not (0.06). A window-length mismatch (4 s real versus 0.18 s simulated) had inflated simulated variance threefold. The data do not constrain the delay model's absolute levels, hence the use of fault *differences*.

Table 13: Classifier two-sample test, real versus simulated healthy windows on held-out cuts (3 random stratified splits, mean ± s.d.; 22 test cuts).

Mean model	Variation model	C2ST balanced accuracy ↓
Linear	mean only	0.946 ± 0.025
Linear	+ between/within-cut variation	0.853 ± 0.022
Kernel ridge	mean only	0.864 ± 0.022
Kernel ridge	+ between/within-cut variation (final)	**0.820 ± 0.036**

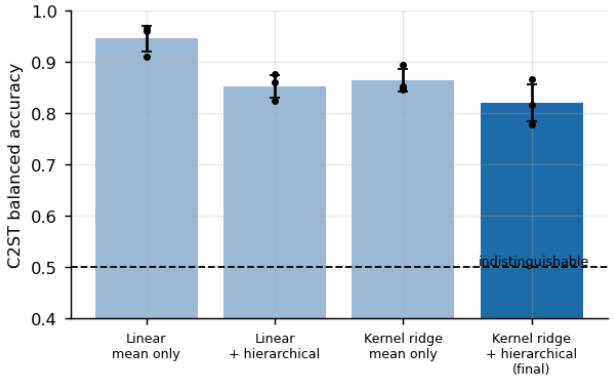


Figure 12: C2ST balanced accuracy of the four baseline variants (dots: individual splits; dashed: indistinguishable).

4.4 Ablation studies

4.4.1 Graph, node construction and fault gain

Table 17 and Figure 18 report m=10 top-1 accuracy over three seeds, each changing the cut split and the generated data (60 draws per fault). Replacing the physical graph by random edges of equal out-degree or by an empty graph changes no algorithm beyond the seed spread (BRCD 0.92 ± 0.01, 0.92 ± 0.02, 0.92 ± 0.00 shared; 0.60 ± 0.05, 0.60 ± 0.05, 0.58 ± 0.06 shifted). Every root-cause node is a graph root, so in our re-implementations the conditional models cannot add information about it and the methods act as marginal shift detectors on the node estimates, consistent with the observation of Lee et al. [14] on BARO and SimpleRCA. Replacing the supervised soft sensors by label-free spectral summaries reduces accuracy to 0.07–0.29 (shared) and 0.07–0.37 (shifted); the simplest method reaches exactly 0.20, consistent with only the controller-measured speed node remaining informative (not tested directly). Under shift, a threefold larger fault gain lowers accuracy (BRCD $0.60 \rightarrow 0.27$, Smooth Traversal $0.71 \rightarrow 0.28$) because soft sensors trained on nominal parameters mis-estimate larger, distorted effects. The benchmark therefore measures supervised estimation of the root-cause nodes and its sensitivity to generator mismatch, not causal reasoning.

4.4.2 Feature normalisation for transfer between machines

Normalising features by each machine's healthy baseline did not raise AUROC on unseen machines (gradient boosting: 0.990 raw, 0.955 machine-relative, 0.986 tool-relative) and increased the wear-estimation error (MAE of VB 9.2, 17.7, 11.1 μm). Domain adaptation does not improve on gradient boosting, and tool-relative features make DANN unstable (machine 1: 0.79 ± 0.13; Table 10, Figure 9).

Table 14: Shift of each sensor node in healthy standard deviations: real defective tool D* (95% bootstrap interval over cuts) versus five simulated faults. Red (blue) cells: shift ≥ 2 (≤ -2) s.d.

	Real tool D*	Simulated fault				
Sensor node [s.d.]	[95% CI]	Stiff.	Damp.	Wear	Depth	RPM
log acc. rms	+1.9 [+1.4, +2.5]	+2.9	+2.9	+1.9	+2.9	+3.0
mic. level	+3.6 [+3.2, +4.1]	+2.3	+2.4	+1.4	+2.3	+2.4
non-sync. index (acc)	-3.7 [-4.3, -3.1]	-0.0	-0.3	-0.3	-0.2	+0.3
non-sync. index (mic)	-1.2 [-1.5, -0.9]	+0.1	-0.2	-0.2	-0.2	+0.5
acc. 50–600 Hz	+3.9 [+3.5, +4.2]	-0.0	+0.0	-0.0	-0.0	+0.0
acc. 1–3.2 kHz	+6.3 [+5.6, +7.1]	+1.8	+1.2	+1.0	+1.3	+1.5
acc. >3.2 kHz	+1.9 [+1.4, +2.3]	+1.5	+1.3	+1.5	+2.1	+2.1
mic. 50–600 Hz	+4.0 [+3.3, +4.7]	-0.0	+0.0	+0.0	-0.0	+0.1
mic. >3.2 kHz	+1.8 [+1.2, +2.3]	+0.2	+0.2	+0.1	+0.3	+0.3
Cosine with real shift	–	0.62	0.60	0.62	0.59	0.55

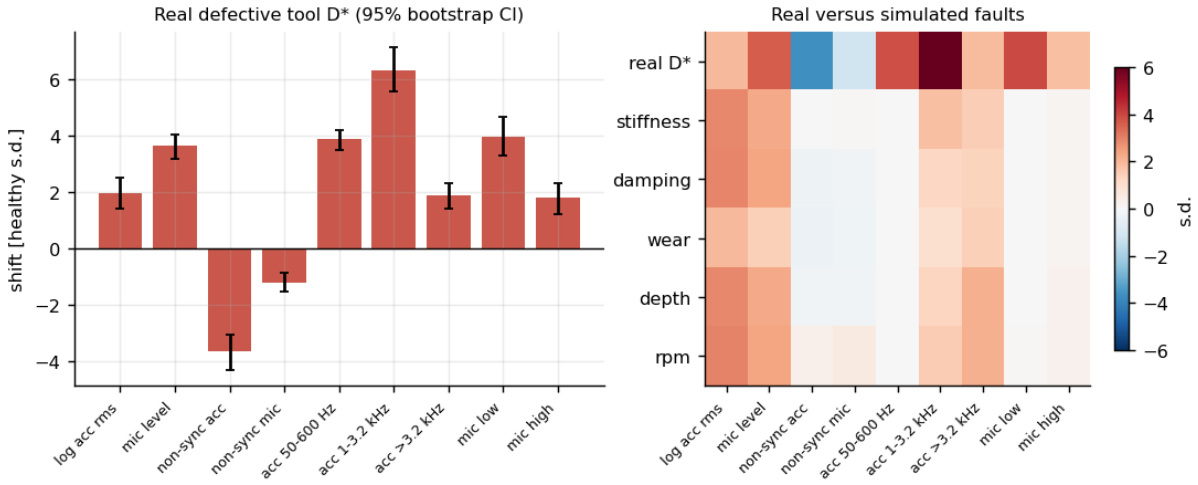


Figure 13: Left: real defective-tool shift with 95% bootstrap intervals. Right: real versus simulated fault shifts on the same scale. Simulated faults raise broadband levels but do not move the low-frequency bands or the non-synchronous indices.

4.4.3 Wear threshold and fine-tuning

Because heavy wear may be too easy, we repeated the label-free comparison at VB ≥ 60 and ≥ 40 μm (thresholds fixed before running; all three are reported). The comparator is the resultant-force rule on the same tool-relative x, y features (Table 18, Figure 19). Averaged over machines, the simulation-only model is higher than the rule at every threshold, but the gain comes from machine 3, where the rule is below chance (0.42, 0.40 and 0.43) and simulation reaches 0.57–0.78; on machines 1 and 2 the rule is as good as or better than simulation in five of six cases (machine 1 at 60 μm is tied). The evidence for a label-free benefit of simulation is therefore limited to one of three machines, with the sign of the one-feature rule on that machine unexplained. Fine-tuning from the simulated solution with k real samples was not better than training from scratch at any k and any threshold (Figure 20; at k=2, λ=1: 0.83 vs. 0.85 at 100 μm, 0.90 vs. 0.94 at 60 μm, 0.87 vs. 0.97 at 40 μm). On this easy task a few real labels already suffice.

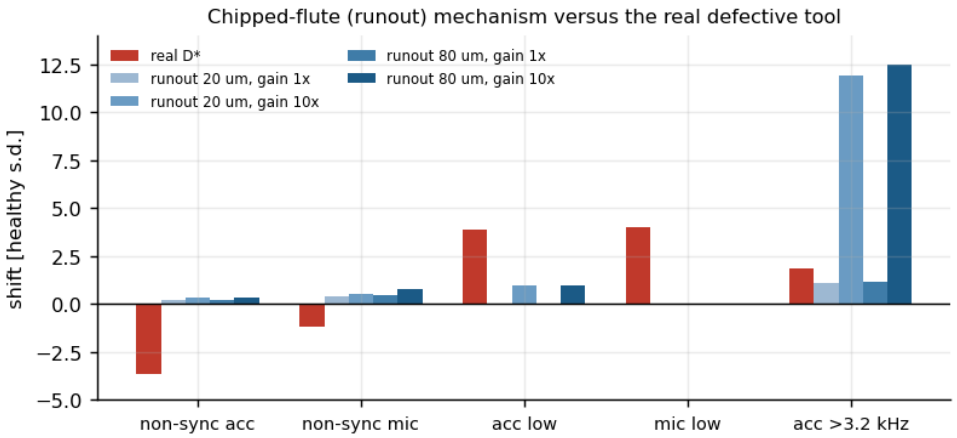


Figure 14: Chipped-flute test: shifts of five discriminating nodes for the real defective tool (red) and for simulated effective runout of 20 and 80 μm at fault gains 1× and 10×.

Table 15: Top-1 accuracy ↑ of classifiers on simulated data versus fault gain relative to the calibrated value.

Classifier	gain ×0.3 ↑	×1 (calibrated) ↑	×3 ↑
Gradient boosting	**0.698**	**0.882**	**0.943**
Random forest	0.674	0.855	0.922
QDA (Bayes)	0.545	0.769	0.847
Logistic regression	0.535	0.729	0.834

5 Conclusion, Limitations and Future Work

5.1 Conclusion

We asked how far simulation can take root cause diagnosis in milling and measured it against real data in three places. Simulated causal-RCA accuracy depends on how diagnostic nodes are built and falls from 0.80–0.98 to 0.35–0.71 under moderate generator mismatch, while the causal graph has no effect. The simulated healthy baseline and the signature of a real defective tool remain different from real data, and the nominal simulator operates above its stability limit, which the effect-size gains silently compensate. On a public wear dataset simulation-only transfer is below a one-feature rule on two machines and brings no fine-tuning benefit, because the task is too easy to show one. Simulation is necessary in principle for causes without real labels, but its value was not demonstrated on any real task we could test; the contribution of this paper is the decomposition of the simulation gap, the measurements and the open pipeline needed to test it on real multi-cause data.

5.2 Limitations

- **No real multi-cause data.** Results are for one tool and material (MSM subset), one defective-tool comparison on a different tool, and force-only wear data (LUH); root-cause ranking has not been validated on real cuts.
- **Simulator.** The nominal simulator is above its stability limit at every real cutting condition; fault severity ranges, gains, nominal physical parameters and the causal graph are assumptions; neither standard fault models nor a chipped-flute runout reproduce the defective-tool

Table 16: Six-class classification on simulated data, 16 held-out cuts (chance 0.167).

Classifier	Top-1 ↑	Top-3 ↑	Macro-F1 ↑	$\text{AUROC}_{\text{fault vs. normal}}$ ↑
Gradient boosting	**0.886**	**0.990**	**0.885**	**1.000**
Random forest	0.870	**0.990**	0.868	0.998
MLP	0.807	0.987	0.804	0.994
QDA (Bayes)	0.787	0.978	0.779	0.992
Logistic regression	0.726	0.966	0.722	0.990
kNN	0.722	0.922	0.717	0.983
Gaussian Naive Bayes	0.579	0.912	0.554	0.961

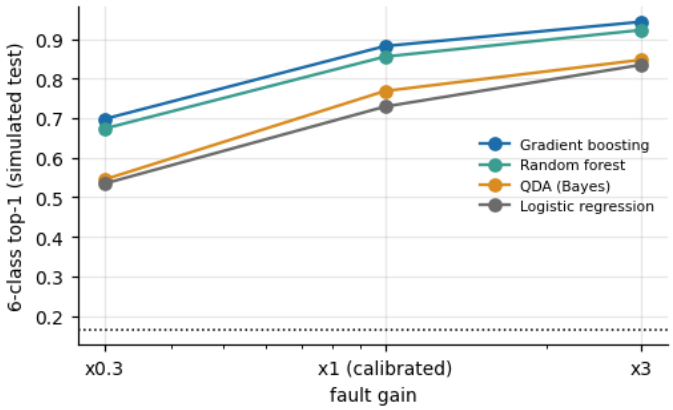


Figure 15: Simulated six-class accuracy as a function of the assumed fault gain.

signature; the healthy baseline is distinguishable from real data (balanced accuracy 0.82–0.95 over the four variants).

- **RCA algorithms.** The algorithms are simplified re-implementations with unsourced constants, so the finding on the causal graph does not transfer to the original algorithms; the main causal tables use one split and seed, and only the ablation reports three seeds.
- **Data.** The MSM microphone and accelerometer show a machine line near 1.5 kHz that contaminates the 1–3 kHz features; the dynamics-distance test is a weak proxy with nine targets; the FRF importer is tested only on synthetic transfer functions; meta domain generalisation is not implemented as a baseline.

5.3 Future work

The decisive experiment needs real multi-cause data. We will collect cuts with designed causes (clamp stiffness, overhang, wear, depth or speed outside the stable range, imbalance, chipped flute) on at least two machines with hammer-test FRFs and force-coefficient calibration cuts, and evaluate by leave-one-machine-out and leave-one-configuration-out with simulation only (k=0), simulation plus k real samples, real only, and real data from other configurations with and without FRF-anchored simulation (Table 19, Figure 21; protocol in `paper/DATA_PROTOCOL.md`). Baselines will include standard classifiers, domain-adversarial transfer [23], meta domain generalisation [16], rule thresholds, FRF-based stability checks, and faithful implementations of BRCD, RCD and RCG. Decision gates: if the simulator with measured FRFs does not reproduce the measured stability

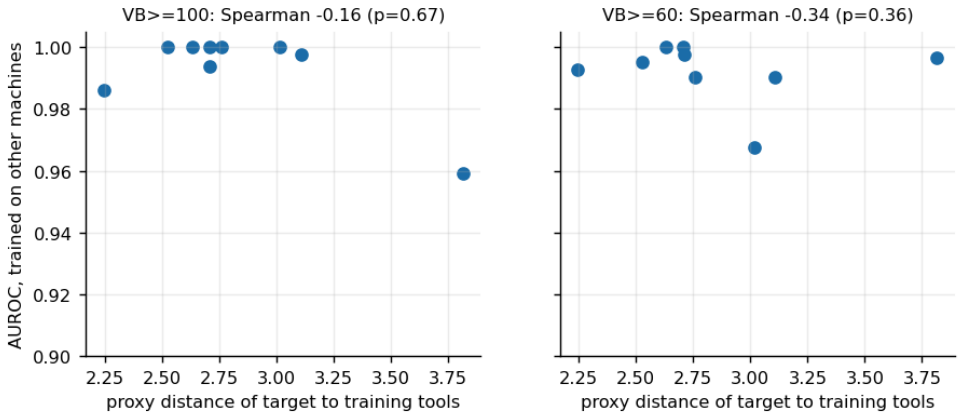


Figure 16: Real-data transfer to an unseen machine versus the spectral proxy distance of the target tool to the training tools (nine targets).

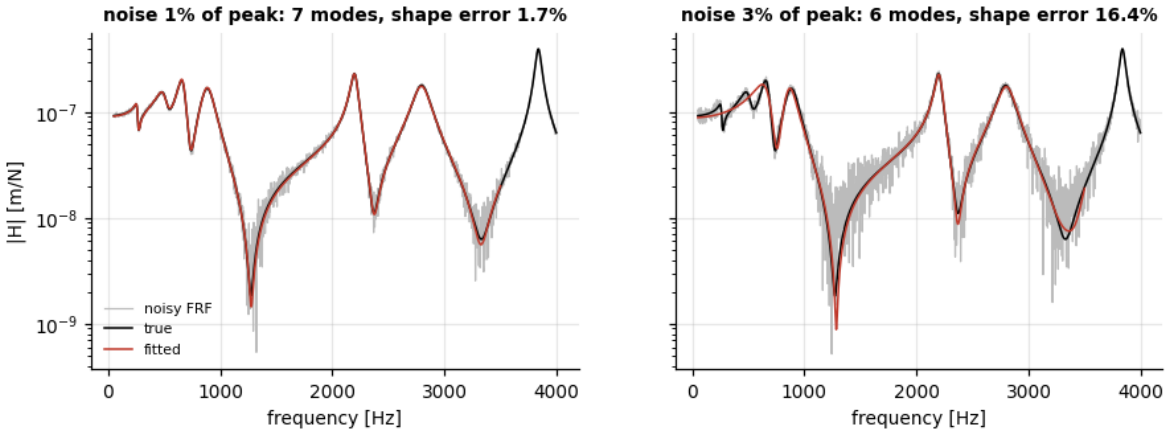


Figure 17: FRF import on a synthetic transfer function with 1% and 3% noise: noisy input, true and fitted receptance magnitude.

boundary within an agreed tolerance, the model is fixed before more data are collected; if no benefit at small k appears, the contribution is positioned as zero-shot transfer, dataset and gap analysis. Further technical work: a stable-regime parameter set learned from measured FRFs, per-tooth force asymmetry and process damping for the defective-tool signature, calibration and abstention for deployment.

Availability

Code. `https://github.com/danghaidang04/RCA-for-Chatter-Simulation` (branch `cutting-sim-experiments`); `REPRODUCE.md` gives environment, data download, run order, seeds and expected outputs; every table and figure is generated from the result files by `paper/make_tables.py` and `paper/make_figures.py`. Licence: MIT (to be confirmed with the university). Archival DOI: to be assigned (Zenodo) at submission.

Table 17: Ablation at m=10: top-1 accuracy ↑ (mean ± s.d. over 3 seeds); chance = 0.07.

Variant	BRCD	RCD	RCG	Smooth	BARO	Simple
Same generator						
Supervised nodes, physics graph (main)	0.92 ± 0.01	0.81 ± 0.03	0.92 ± 0.01	0.99 ± 0.00	0.91 ± 0.07	0.54 ± 0.00
Supervised nodes, shuffled graph	0.92 ± 0.02	0.81 ± 0.03	0.92 ± 0.01	0.99 ± 0.00	0.91 ± 0.07	0.54 ± 0.00
Supervised nodes, no edges	0.92 ± 0.00	0.81 ± 0.02	0.92 ± 0.00	0.99 ± 0.00	0.91 ± 0.07	0.54 ± 0.00
Label-free nodes, physics graph	0.09 ± 0.01	0.10 ± 0.04	0.07 ± 0.01	0.29 ± 0.08	0.29 ± 0.08	0.20 ± 0.00
Supervised, physics graph, gain ×0.3	0.85 ± 0.02	0.78 ± 0.04	0.85 ± 0.02	0.88 ± 0.05	0.65 ± 0.18	0.51 ± 0.01
Supervised, physics graph, gain ×3	0.85 ± 0.02	0.76 ± 0.05	0.84 ± 0.02	0.99 ± 0.01	0.97 ± 0.01	0.53 ± 0.04
Shifted generator parameters						
Supervised nodes, physics graph (main)	0.60 ± 0.05	0.43 ± 0.05	0.60 ± 0.07	0.71 ± 0.07	0.39 ± 0.05	0.45 ± 0.01
Supervised nodes, shuffled graph	0.60 ± 0.05	0.42 ± 0.05	0.59 ± 0.06	0.71 ± 0.07	0.40 ± 0.05	0.45 ± 0.01
Supervised nodes, no edges	0.58 ± 0.06	0.42 ± 0.05	0.58 ± 0.06	0.70 ± 0.06	0.39 ± 0.06	0.45 ± 0.01
Label-free nodes, physics graph	0.09 ± 0.01	0.14 ± 0.04	0.07 ± 0.01	0.27 ± 0.03	0.37 ± 0.02	0.20 ± 0.00
Supervised, physics graph, gain ×0.3	0.54 ± 0.03	0.45 ± 0.05	0.53 ± 0.03	0.59 ± 0.03	0.48 ± 0.02	0.42 ± 0.02
Supervised, physics graph, gain ×3	0.27 ± 0.07	0.22 ± 0.01	0.26 ± 0.07	0.28 ± 0.09	0.22 ± 0.01	0.32 ± 0.07

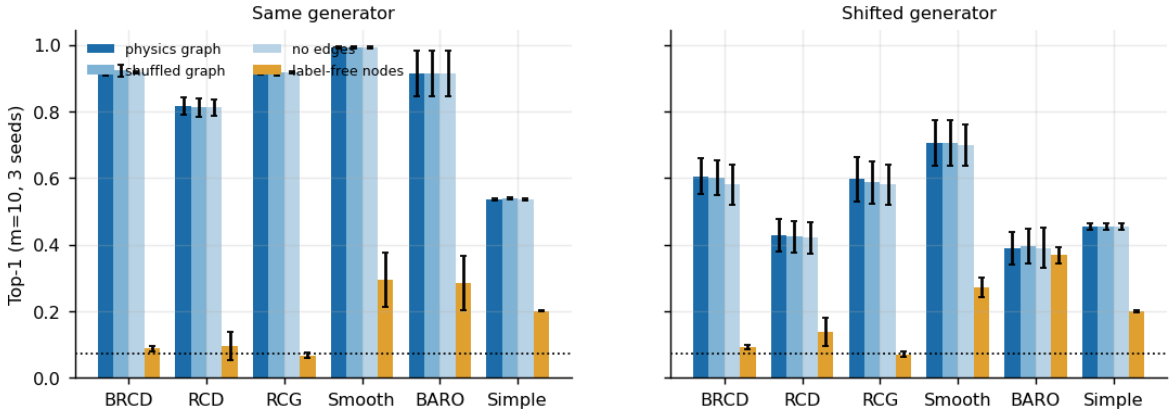


Figure 18: Ablation at m=10 (mean ± s.d. over 3 seeds): the graph does not matter; node construction does.

Data. MSM [13] (Kaggle, DOI 10.34740/kaggle/ds/8392825, CC BY 4.0) and LUH [5] are obtained from their original sources and are not redistributed; derived features and all result files are in the repository.

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Table 18: Label-free wear detection on unseen machines, AUROC ↑, by wear threshold. Rule: mean resultant force on tool-relative x, y features.

VB threshold [μm]	Method	M1 ↑	M2 ↑	M3 ↑	Mean ↑
≥ 100	One-feature rule	0.853	0.936	0.433	0.740
	Simulation only	0.887	0.807	0.571	**0.755**
≥ 60	One-feature rule	0.927	0.870	0.415	0.737
	Simulation only	0.925	0.809	0.759	**0.831**
≥ 40	One-feature rule	0.994	0.941	0.398	0.777
	Simulation only	0.827	0.893	0.782	**0.834**

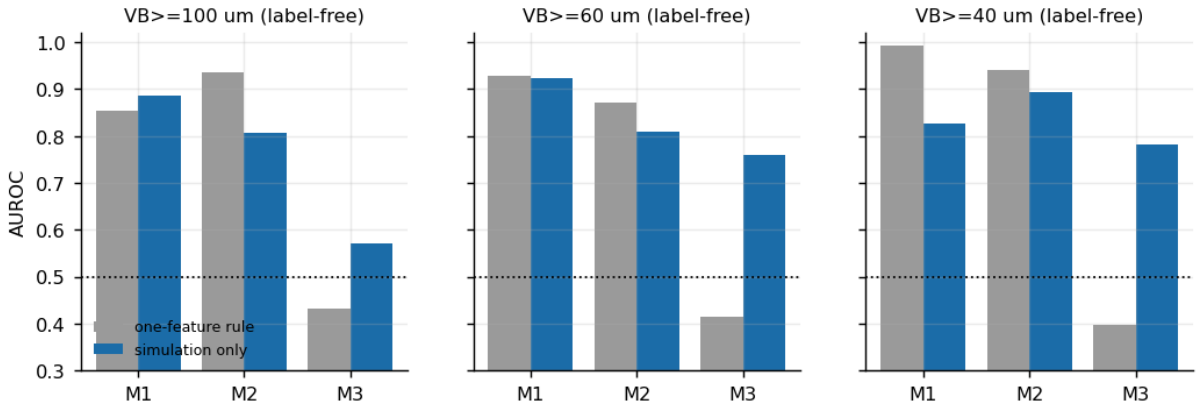


Figure 19: Label-free AUROC of the one-feature rule and the simulation-only model per machine and wear threshold.

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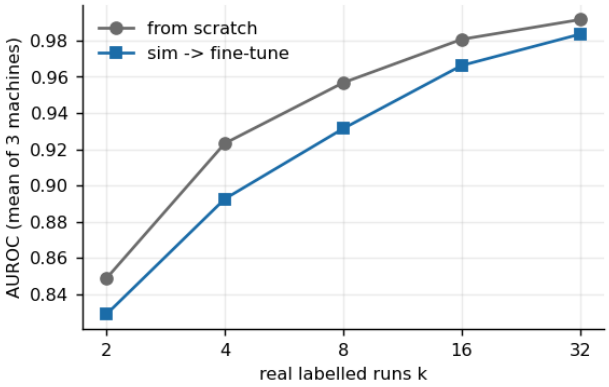


Figure 20: AUROC versus the number k of real labelled runs (VB ≥ 100 μm; mean of three machines).

Table 19: PLACEHOLDER (not real data). Layout of the planned leave-one-configuration-out evaluation on real multi-cause cuts: top-1 ↑ and expected calibration error (ECE) ↓ per training mode, k=8 real cuts.

Training mode	Top-1 ↑	Top-3 ↑	Macro-F1 ↑	ECE ↓
(a) simulation only (k=0)	0.46	0.78	0.41	0.19
(b) simulation + fine-tune (k=8)	**0.70**	**0.93**	**0.66**	**0.08**
(c) real only (k=8)	0.58	0.84	0.52	0.14
(d) real, other configurations, no simulation	0.52	0.81	0.47	0.12
(e) (d) + FRF-anchored simulation	0.66	0.90	0.61	0.10

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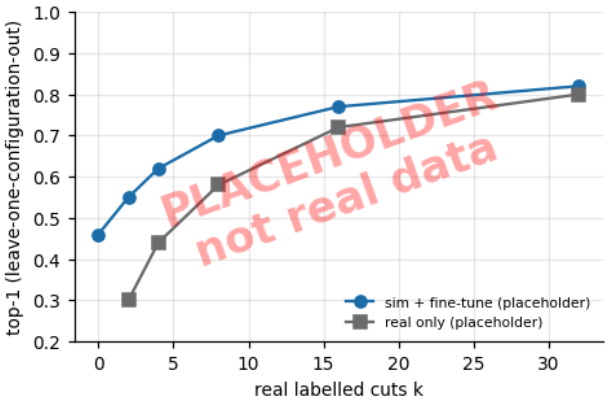


Figure 21: PLACEHOLDER (not real data). Planned learning curve: top-1 versus the number k of real labelled cuts, with and without simulation pre-training.

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A Parameter provenance

B Nominal simulator parameters

C Methodological pitfalls found along the way

1. Plain accuracy of a classifier two-sample test with unequal sizes has a chance level above 0.5; use balanced accuracy.

2. Real and simulated features must use windows of identical length; a 20-fold mismatch inflated variance threefold.

Table 20: Every simulator parameter, its tier (fixed / range / learned / assumed) and verification status.

Parameter	Tier	Verification
tool diameter, flutes, flute length, overall length	FIXED	verified (read from catalog table p.5)
helix angle β	FIXED	medium: from series feature list (p.3); not stated per part number
tool substrate/coating	FIXED	medium
N, ae, feed F, direction	FIXED	verified (N and F equal mtc Spindle_Speed/Feed_Rate, ratio 1.000)
axial depth ap = engaged depth	FIXED	assumption (not measured)
workpiece AA 6061; wet cutting (coolant flood)	FIXED	verified (stated in paper)
sampling: accelerometer 51.2 kHz, audio 48 kHz	FIXED	verified (stated in paper)
analysis window 0.3 s	FIXED	design choice
model structure: regenerative DDE, shear+edge force, ZOA lobes	FIXED	model structure
Kt, Kr, Kte, Kre (AA 6061, this tool, wet)	LEARNED	NO usable source (rejected: steel-like values; dry-cutting AA6061 value unverified)
tool-tip FRF: fn_x, fn_y, ζ, modal stiffness k	LEARNED	NO source for this tool/holder/stick-out (Altintas Ex.2 FRF rejected: different machine and tool)
spindle imbalance U	LEARNED	NO source (rotor mass, balance grade unknown)
sensor gains, noise floors, rpm-dependence	LEARNED	NO source (sensors uncalibrated: mic dBFS, accelerometer mm/s^2)
runout; between-cut variability σ (Kt, runout, roughness)	LEARNED	NO source
chip-thickness cap 0.3 mm; roughness noise base 0.02 μm	ASSUMED	own assumption

3. Out-of-sample residuals are needed to estimate between-cut variation when the regression can interpolate each cut.

4. Distribution matching alone did not identify the fault gain; effect-size matching was used.

5. Nodes trained on the generator's labels make a causal benchmark circular.

6. Force-axis sign conventions differ between machines; a signed mean force put zero-shot transfer below chance.

7. Plotting raw simulator output early would have exposed the stability problem of Section 4.3.2; check the operating regime of a simulator against the stability limit before using it.

Table 21: Nominal values and numerical search bounds of the 16 simulator parameter groups (`cutting_sim/model_v4.py`). The bounds are numerical, not claims from the literature. Fixed quantities: 2 flutes, D=6.35 mm, β=30°, a_p=38.1 mm, N_z=8, Δt=1/25 600 s, $h_{\max}$=0.3 mm.

Parameter	Nominal	Unit	Numerical search bounds
g_{acc}	2.48e-03	–	6.14e-06 to 7.39e+00
acc. noise floor	0.018	g	0.000 to 1.000
g_{mic}	1.00	–	0.00 to 403.43
mic. noise floor	0.050	Pa	0.000 to 1.000
natural freq. f_x	1500	Hz	500 to 4000
natural freq. f_y	1500	Hz	500 to 4000
damping ratio ζ	0.030	–	0.005 to 0.150
modal stiffness k	3.0e+06	N/m	5.0e+05 to 2.0e+07
tangential coeff. K_t	7.0e+08	N/m^2	2.0e+08 to 2.0e+09
ratio K_r/K_t	0.30	–	0.10 to 1.50
edge coeff. K_{te}	2.0e+04	N/m	1.0e+03 to 1.0e+05
edge coeff. K_{re}	2.0e+04	N/m	1.0e+03 to 1.0e+05
imbalance U	2.3e-06	kg m	2.1e-09 to 2.5e-03
acc. noise rpm exponent	1.0	–	-2.0 to 8.0
mic. noise rpm exponent	1.0	–	-2.0 to 8.0
runout	5.0	μm	1.0 to 30.0